

Input image I

CLIP ViT-B/16 Visual encoder

patches

ROI pooling

YOLO detector

Boxes B

Linear projection

V

Object alignment (A)

Two cascaded stages

Semantic banks (see bank detail)

Semantic banks Nouns / verbs

Semantic aggregation

Semantic aggregation

P_n

V_o

P_v

Relation alignment (A)

V_r

Image-specific object–relation prompts

Image I

UPT detector

Object / relation phrases

CLIP text encoder

ROI matching Class + IoU

YOLO boxes B

Region-linked prompt alignment (detail B)

Slot MLPs

Q

Masked cross-attention

K,V

Projection

V_o'

Q

Masked cross-attention

K,V

Projection

V_r'

Candidate scoring and selection

Decoder likelihood

CLIPScore

Object consistency

Weighted selection

Final caption

$S = 0.3D + 0.4C + 0.3O$

Five candidate captions

Beam search K = 5

Transformer decoder ×3

Token prefix